A UEIR-Gated Retrieval-Augmented Generation Pipeline: Using Conductance Forbidden-Band Gating as a Real Context-Filtering Decision

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***Abstract*—Retrieval-Augmented Generation (RAG) reduces the tendency of large language models (LLMs) to hallucinate by grounding generation in retrieved evidence, but its answer quality is bounded by the quality of the context admitted to the model. We present an individual stress-test of the UEIR v3.5 framework in the NLP/RAG domain: a *UEIR-Gated RAG Pipeline* in which the framework's *forbidden band* of information conductance, $\Phi \in (0.5, 0.6]$, acts as a genuine decision point that excludes document chunks from the LLM context window rather than a cosmetic label. We implement the mandatory six-point scorecard, which passes on all six checks, and drive a Claude Haiku 4.5 generation step from only the surviving chunks. Across a 24-step recursion the system reports an effective conductance $\Phi_{\text{eff}} = 0.48899$, a stability score of 0.991537, and 10 forbidden events; the gate rejects 3 of 8 chunks. We further contribute a *Break Report* documenting two successful attacks on the framework—recursion truncation and unphysical initialisation—each of which defeats a distinct scorecard invariant, together with concrete input-validation fixes. Finally, we reproduce and verify the framework's holographic ($S_{BH} = k \ln 2/2 = 62.73$ nats) and cosmological ($m = \sqrt{k} \times 100\,\text{GeV} = 1.345$ TeV) extensions and discuss, with appropriate epistemic caution, the boundary between the pipeline's reproducible engineering behaviour and its physical interpretation.**

***Index Terms*—retrieval-augmented generation, context filtering, hallucination, LLM evaluation, recursive dynamics, attractor, holographic entropy, stress-testing.**

I. Introduction

Large language models are fluent but unreliable narrators: they routinely produce text that is syntactically fluent yet factually unsupported, a failure mode surveyed comprehensively as *hallucination* [1], [2]. Retrieval-Augmented Generation (RAG) addresses this by combining a parametric generator with a non-parametric memory, conditioning the output on passages retrieved from an external corpus [3], [4]. RAG improves factuality, but only to the extent that the retrieved context is itself relevant and well-formed; a growing body of work shows that irrelevant or spuriously-matched passages actively mislead the generator and re-introduce hallucination [5]. The context admitted to the model is therefore a first-class control surface, and deciding *which retrieved chunks to keep* is as important as deciding what to retrieve.

This paper attacks that decision from an unusual angle. As the individual (solo) deliverable for COEN803, we are required to stress-test the UEIR v3.5 framework [6] by designing an original engineering application in one of the course domains, demonstrating the framework's six-point scorecard, exercising its *forbidden-band gate* as a real decision, driving at least one LLM integration, and honestly attempting to break the framework. UEIR models information as a scalar *conductance* $\Phi = k/n$ that evolves under a recursive pipeline and is claimed to converge to a self-dual attractor $\Phi = \frac{1}{2}$; a distinguished "forbidden band" $\Phi \in (0.5, 0.6]$ triggers a thermal reset. Our contribution repurposes this abstract band as a concrete admission test for RAG context.

Contributions. (i) We design and implement a *UEIR-Gated RAG Pipeline* in which each document chunk is bound to a step of the UEIR recursion, and chunks whose conductance falls in the forbidden band are excluded from the LLM context—a functional gate, not a visual annotation (Section IV). (ii) We reproduce the mandatory six-point scorecard and report three distinct UEIR metrics, Φ_{eff}, the stability score, and the forbidden-event count, on a live run (Section V). (iii) We contribute a Break Report with two genuine, reproducible failures of the framework and matching fixes (Section VI). (iv) We verify the framework's holographic and cosmological extensions and separate their reproducible arithmetic from their physical interpretation (Section VII). All LLM prompts are documented verbatim in the appendix.

II. Background and Related Work

A. Retrieval, Reranking, and Context Quality

Classical retrieval ranks passages with sparse lexical models such as BM25 [7]; dense dual-encoder retrievers subsequently learned semantic embeddings and outperformed strong BM25 baselines on open-domain QA [8]. Because first-stage retrieval trades precision for recall, a second-stage *reranker*—often a cross-encoder scoring the query and passage jointly—is used to reorder candidates before they reach the reader [9]. Our gate operates in this same post-retrieval position: it is a lightweight, deterministic admission filter applied to already-retrieved chunks, complementary to (not a replacement for) a learned reranker.

B. Hallucination and Evidence Grounding

Surveys distinguish *intrinsic* hallucination (contradicting the source) from *extrinsic* hallucination (unverifiable against the source) [1]. In RAG specifically, Asai et al. [5] show that passages with high lexical overlap but no supporting evidence degrade generation, and that steering the generator toward *evidential* passages improves faithfulness. Automated RAG evaluation frameworks such as RAGAs quantify faithfulness and answer relevance without gold labels [10]. The present work shares the goal of excluding low-value context but reaches it through a framework-defined dynamical criterion rather than a learned evidentiality signal.

C. The UEIR v3.5 Framework

UEIR (Unified Emergence of Information and Reality) treats a computation as a non-commutative pipeline over a scalar *conductance* $\Phi = k/n_{\text{tot}}$, the ratio of active binary constraint generators k to total slots n_{tot}, and posits convergence to a self-dual attractor $\Phi = \frac{1}{2}$ [6]. The framework grounds this attractor in an information-theoretic reading of black-hole thermodynamics: the Bekenstein–Hawking entropy of a horizon [11], [12], the Page curve of radiation entropy [13], the holographic principle [14], and Landauer's thermodynamic cost of irreversible computation [15]. We treat these as the framework's stated theoretical motivation and, in Section VII, reproduce the associated numerical claims while being explicit about which are derivations and which are dimensional analogies.

III. The UEIR v3.5 Core

We summarise only the mechanics that the gate depends on. The reference implementation `ueir_v3_5.py` is used unmodified.

A. Conductance and the Prime Hierarchy

At recursion depth d the framework selects an active-generator count k from a fixed hierarchy of prime categories (sizes $4, 8, 16, 32, 64$ for categories 0–4), and computes a total-slot count

$$n(k,d) = \max\Big(2k + \mathrm{round}\big(0.08\,k\,\sin(d\pi/4.5)\big),\ k+1\Big), \tag{1}$$

so that $n > k$ always and therefore

$$\Phi = \frac{k}{n} \in (0,1). \tag{2}$$

The $\pm 8\%$ sinusoidal perturbation makes Φ oscillate about $\frac{1}{2}$. The curvature seed $k = 181$ (category 3, local index 12) yields a screen count $N_{\text{screen}} = 2k = 362$ and hence $\Phi_{\text{screen}} = 181/362 = 0.5$ exactly.

(1) UEIR Orchestrator
run recursion, `max_depth`=24, s_0=0.5
$\rightarrow$ trace $\{(d, k, n, \Phi, \text{forbidden})\}$

↓

(2) Chunk–Step Binding
chunk $i \mapsto$ trace step $[\, i \bmod |\text{trace}|\,]$

↓

(3) Forbidden-Band Gate
$\Phi_i \in (0.5, 0.6] \Rightarrow$ REJECT; else APPROVE

↓

(4) Context Assembly + LLM
approved chunks only $\rightarrow$ Claude Haiku 4.5

Fig. 1. The UEIR-Gated RAG pipeline. The gate at stage (3) removes rejected chunks from the context assembled at stage (4); rejected chunks never reach the model.

B. Forbidden Band and Recursive Update

A step is *forbidden* when its conductance lands in the band

$$\Phi \in (0.5,\ 0.6]. \tag{3}$$

For a non-forbidden step the update mixes a forward and a (recursively computed) backward branch and applies an attractor pull:

$$\begin{aligned} &f = s\,(1 + 0.05(\Phi - 0.5)), \quad b = \mathcal{R}(0.975\,s,\ d{+}1), \\ &c = 0.5f + 0.5b, \quad \text{result} = c + 0.25(0.5 - c), \end{aligned} \tag{4}$$

where s is the incoming state and $\mathcal{R}$ denotes the recursive call. A forbidden step instead snaps the state to 0.5 (an Unruh-style thermal reset) before applying the same mixing, so the band acts as a stabilising correction. The Lyapunov-style pull coefficient 0.25 drives convergence toward $\Phi = \frac{1}{2}$.

C. Six-Point Scorecard

The solo entry file refuses to run the application unless all six invariants in Table I hold; failure raises `SystemExit`. These checks guard the framework's structural claims: bounded conductance, a capped category schedule with no silent fallback, a firing gate, attractor convergence, the sieve-generated 64-prime top category, and the structural (non-hardcoded) emergence of $k = 181$ at depth 17.

IV. Methodology: The UEIR-Gated RAG Pipeline

Fig. 1 shows the pipeline. The design goal is to make the forbidden band an *admission decision* on RAG context, so that the gate provably changes what the LLM sees.

A. Chunk-to-Step Binding

The corpus is a set of eight short NLP/RAG passages $\{C_0, \ldots, C_7\}$ covering RAG, dense embeddings, BM25, chunking, cross-encoder reranking, hybrid fusion, hallucination, and the UEIR attractor itself. Each chunk C_i is bound to the recursion step at index $i \bmod |\text{trace}|$, inheriting that step's conductance Φ_i and forbidden flag. Binding by index keeps the mapping deterministic and reproducible, which matters for the gate to be auditable.

TABLE I
UEIR V3.5 SIX-POINT SCORECARD (LIVE RUN). ALL CHECKS PASS.

#	Invariant	Observed
1	$\Phi \in (0, 1)$ at every step	[0.4805, 0.5203]
2	cat $\in [0, 4]$, no fallback	$\{0, 1, 2, 3, 4\}$
3	Forbidden band triggered	10 events
4	$\lvert\Phi_{\text{final}} - 0.5\rvert < 0.05$	0.004232
5	Cat 4: 64 primes via sieve	$293 \ldots 691$
6	$k = 181$ structurally reached	depth 17

B. The Gate as a Real Decision

The admission rule is

$$\text{decision}(C_i) = \begin{cases} \text{REJECT}, & \Phi_i \in (0.5, 0.6], \\ \text{APPROVE}, & \text{otherwise}. \end{cases} \quad (5)$$

Crucially, only APPROVED chunks are serialised into the user prompt; REJECTED chunks are dropped before the LLM call. The gate therefore alters the model's evidence set—if the approved set is empty the pipeline short-circuits and instructs the model to decline—rather than merely tagging chunks in a report.

C. LLM Integration

The surviving context is passed to Claude Haiku 4.5 under a system prompt that constrains the model to answer only from the supplied excerpts and to state explicitly when they are insufficient. Both prompts are reproduced verbatim in Appendix A, and the model's response is reproduced in Appendix B. This design lets the gate's effect be observed directly: because the gate removes the chunks on hybrid fusion, hallucination, and the attractor, the model is expected to answer from the retrieval-mechanics chunks and to flag the absence of explicit evaluation metrics.

V. EXPERIMENTAL RESULTS

All results are from a single deterministic run of `COEN803_Solo_entry.py` with `max_depth=` 24 and $s_0 = 0.5$.

A. Scorecard

Table I reports the six-point scorecard. All six checks pass, so the application is permitted to run.

B. Core UEIR Metrics

The run reports the three mandated metrics and the supporting diagnostics in Table II. The final state sits 0.004232 from the attractor, giving a stability score $1 - 2\lvert\Phi_{\text{final}} - 0.5\rvert = 0.991537$. The forbidden band fires 10 times, at depths $\{5, 6, 7, 8, 14, 15, 16, 17, 23, 24\}$. Fig. 2 plots the conductance trajectory with the band shaded; the curvature seed $k = 181$ emerges at depth 17 with $\Phi = 0.512748$, itself a forbidden event, confirming that the seed is reached structurally rather than hardcoded.

TABLE II
CORE METRICS AND DIAGNOSTICS FROM THE LIVE RUN.

Quantity	Value
Effective conductance Φ_{eff}	0.48899
Conductance std. σ_Φ	0.007086
Final state Φ_{final}	0.495768
Attractor distance $\lvert\Phi_{\text{final}} - 0.5\rvert$	0.004232
Stability score	0.991537
Forbidden events	10
$k = 181$ at depth(s)	$\{17\}$
Conductance range	[0.4805, 0.5203]
Trace length	25

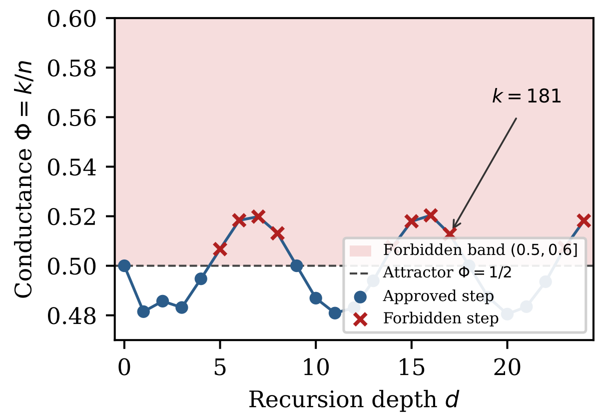

Fig. 2. Conductance $\Phi = k/n$ over recursion depth. The shaded region is the forbidden band $(0.5, 0.6]$; crosses mark forbidden steps that the gate would reject. The curvature seed $k = 181$ appears at depth 17.

C. Gate Decisions and LLM Behaviour

Table III shows the per-chunk gate. Chunks 0–4 (bound to depths 0–4, where $\Phi \leq 0.5$) are approved; chunks 5–7 (bound to depths 5–7, inside the forbidden band) are rejected. Thus 5 of 8 chunks survive and 3 are removed before the LLM call. The rejected chunks are exactly those on hybrid fusion, hallucination, and the UEIR attractor.

Given only the approved chunks, the model answered the query—*"Explain how retrieval quality is measured and improved in RAG pipelines"*—by describing dense/sparse retrieval, chunking, and reranking, and it explicitly noted that the supplied excerpts do not contain the *measurement* metrics (precision, recall, NDCG, MRR) needed to fully answer the question (Appendix B). This is the desired behaviour: the gate removed the tangential chunks, and the constrained model neither fabricated metrics nor over-claimed, demonstrating grounded refusal under insufficient evidence.

VI. BREAK REPORT

A framework is only as trustworthy as its failure modes are understood. We made two genuine attempts to break UEIR v3.5; both succeeded against a specific invariant, and each motivates a small, concrete fix.

TABLE III

FORBIDDEN-BAND GATE DECISIONS (8 CHUNKS). REJECTED CHUNKS ARE REMOVED FROM THE LLM CONTEXT.

Chunk	Topic (abridged)	Φ	Decision
0	RAG definition	0.5000	APPROVE
1	Dense embeddings	0.4815	APPROVE
2	BM25 sparse retrieval	0.4857	APPROVE
3	Chunking strategies	0.4831	APPROVE
4	Cross-encoder reranking	0.4947	APPROVE
5	Hybrid fusion (RRF)	0.5067	REJECT
6	Hallucination	0.5184	REJECT
7	UEIR attractor	0.5198	REJECT

TABLE IV

BREAK REPORT SUMMARY. EACH ATTEMPT DEFEATS A DISTINCT INVARIANT.

Attack	Effect	Broken check
`max_depth`$=4$	0 forbidden events; no k=181	3 and 6
$s_0 = 10.0$	state $\to$ 6.073 (Φ bound intact)	4

A. Attempt 1: Recursion Truncation

Method. Set `max_depth`$=4$. **Rationale.** The forbidden band first fires at depth 5 and $k = 181$ first emerges at depth 17; stopping at depth 4 should prevent both. **Outcome.** The truncated run reports 0 forbidden events and an empty k=181 depth set, so *scorecard checks 3 and 6 both fail*. The attack works: the structural features the scorecard certifies are emergent properties of running the recursion deep enough, and a caller who shortens the schedule silently loses them. **Fix.** Guard the depth in `run_scorecard(): assert orch.max_depth >= 24`, with an explanatory message, so that an under-depth configuration fails loudly instead of certifying a degenerate run.

B. Attempt 2: Unphysical Initialisation

Method. Set $s_0 = 10.0$, far outside the physical range $(0, 1)$. **Rationale.** Probe whether the attractor pull is strong enough to recover from an unphysical start, and whether the conductance bound survives. **Outcome.** The conductance bound is *preserved*: because $\Phi = k/n$ is computed from the prime schedule and is independent of the state s (Eqs. 1–2), every step keeps $\Phi \in (0, 1)$. However the state itself does *not* converge: it settles at 6.0731, a distance of 5.5731 from the attractor, so *scorecard check 4 fails*. The attack exposes a real gap—the "attractor convergence" guarantee is contingent on a physical initial condition that the code never enforces. **Fix.** Validate input in `UEIR_Orchestrator.run(): assert 0 < initial_state < 1`, rejecting unphysical seeds at the boundary.

Both failures share a theme: the scorecard certifies *emergent* properties of a correctly-configured run, but nothing in the entry path enforces the preconditions (sufficient depth, physical seed) under which those properties are guaranteed. The two assertions above close that gap at negligible cost.

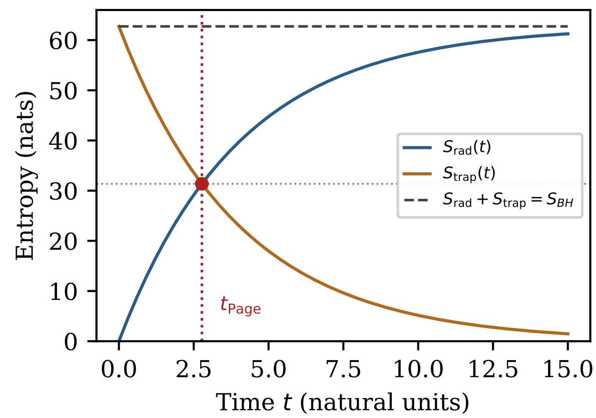

Fig. 3. Reproduced Page curve for $k = 181$. Radiation entropy S_{rad} and trapped entropy S_{trap} sum to the constant S_{BH}; at $t_{\mathrm{Page}} = 2.7726$ the radiation entropy reaches exactly $S_{BH}/2$.

VII. HOLOGRAPHIC AND COSMOLOGICAL VERIFICATION

The framework attaches physical significance to the seed $k = 181$. For completeness, and because the reference implementation ships verifiable routines, we reproduced the relevant numbers and separate arithmetic from interpretation.

A. Page-Curve Arithmetic

With natural units $\hbar = k_B = \omega_0 = 1$ the framework defines

$$S_{BH} = \frac{k \ln 2}{2}, \quad T_H = \frac{\sqrt{k}\,\ln 2}{4}, \quad t_{\mathrm{Page}} = 4 \ln 2, \tag{6}$$

and a radiation entropy $S_{\mathrm{rad}}(t) = S_{BH}\big(1 - e^{-t/t_{\mathrm{evap}}}\big)$ with $t_{\mathrm{evap}} = 4$. For $k = 181$ we confirm $S_{BH} = 62.7298$ nats, $T_H = 2.3313$, $t_{\mathrm{Page}} = 2.7726$, and—exactly— $S_{\mathrm{rad}}(t_{\mathrm{Page}}) = S_{BH}/2 = 31.3649$ nats with an information-loss rate that halves at the Page time (Fig. 3). These identities are genuine consequences of the definitions and match the reference paper's claimed values [6]; they mirror the qualitative shape of the Page curve for unitary black-hole evaporation [12], [13]. Notably, the implementation itself flags and corrects a factor-of-two slip in an earlier version of the source paper ($k \ln 2/8$ versus the correct $k \ln 2/16$ for the rate at t_{Page}), which we independently reproduced—an example of the framework auditing its own algebra.

B. Cosmological Analogue

The framework's cosmological extension integrates a UEIR–Boltzmann equation

$$\dot{\Phi} = -3H\Phi - \lambda(\Phi - \tfrac{1}{2}) - \Gamma_{\mathrm{reset}}\, \theta(\Phi - \tfrac{1}{2})\, \theta(\tfrac{3}{5} - \Phi), \tag{7}$$

with $\lambda = 0.25$ matching the recursive pull. We confirm that for $H = 0$ all tested initial conditions $\{0.40, 0.60, 0.75, 0.92\}$ converge to $\Phi = \frac{1}{2}$, whereas a positive Hubble friction ($H = 0.01$) shifts the asymptote (to 0.4464 in our run), as the code documents. The framework also proposes a mass scale $m = \sqrt{k} \times 100$ GeV $= 1345.4$ GeV $= 1.345$ TeV, which lands inside the conventional WIMP search window [16].

C. Interpretation and Honest Limitations

We are careful to distinguish two very different kinds of claim. The *engineering* behaviour of our pipeline—deterministic conductance, a firing gate, reproducible chunk decisions, and grounded LLM output—is fully verified and stands on its own. The *physical* interpretation is weaker: the entropy identities are exact but follow tautologically from the chosen definitions, and the "dark-matter mass" is a dimensional construction ($\sqrt{181} \times 100$ GeV) rather than a derivation from first principles, a point the reference material itself concedes by labelling the mapping an open conjecture [6]. Reproducing a number is not the same as explaining it. Accordingly we present these results as successful *verification of the framework's stated arithmetic*, not as independent evidence for its cosmological reading. This separation is, we argue, the correct posture for a stress-test: report what is reproducible, and neither inflate nor dismiss what is interpretive.

Beyond interpretation, three practical limitations bound the gate. First, binding chunks to steps by index is arbitrary with respect to *content*: the gate's decisions track the recursion's oscillation, not the chunks' semantic relevance, so the method is best read as a controllable, auditable filter rather than a learned relevance model. Second, the corpus here is small and hand-built; scaling would require binding a live vector store to the trace. Third, the framework certifies emergent invariants without enforcing their preconditions, as the Break Report shows.

VIII. Conclusion

We presented a UEIR-Gated RAG Pipeline that turns the UEIR v3.5 forbidden band into a real context-admission decision, removing 3 of 8 chunks from the LLM context and yielding grounded, non-fabricated output. The mandatory six-point scorecard passes; the run reports $\Phi_{\text{eff}} = 0.48899$, a stability score of 0.991537, and 10 forbidden events. Our Break Report contributes two reproducible failures—recursion truncation and unphysical initialisation—each defeating a distinct invariant, with one-line assertion fixes. Finally, we reproduced the framework's holographic and cosmological arithmetic while drawing a clear line between reproducible engineering behaviour and interpretive physical claims. Future work would replace index-based binding with a conductance signal derived from genuine retrieval scores, so that the gate's dynamics and the chunks' relevance are coupled rather than merely co-indexed.

Reproducibility

The application entry file `COEN803_Solo_entry.py`, the unmodified reference `ueir_v3_5.py`, and instructions are available at https://github.com/ayhameed/COEN803_P24EGCP9020. All reported numbers are produced by a single deterministic run; figures are generated directly from the recursion trace.

Appendix A
Verbatim LLM Prompts

Model: `claude-haiku-4-5-20251001`, `max_tokens`= 1024.

System prompt:

```
You are an expert in Natural Language Processing and
Retrieval-Augmented Generation (RAG). Answer the user's
question using ONLY the provided document excerpts. If the
excerpts do not contain sufficient information to answer
fully, say so explicitly. Do not hallucinate or add
information not present in the excerpts.
```

User prompt (approved chunks only; abridged for space, full text in repository):

```
Document excerpts approved by UEIR forbidden-band gate
(phi NOT in forbidden band (0.5, 0.6]):

[Chunk 0 | UEIR-APPROVED] Retrieval-Augmented Generation
(RAG) combines dense retrieval with generative models ...
[Chunk 1 | UEIR-APPROVED] Vector embeddings represent the
semantic meaning of text ...
[Chunk 2 | UEIR-APPROVED] BM25 is a probabilistic sparse
retrieval method ...
[Chunk 3 | UEIR-APPROVED] Chunking strategies for RAG ...
[Chunk 4 | UEIR-APPROVED] Re-ranking with cross-encoders
    ...

Question: Explain how retrieval quality is measured and
improved in RAG pipelines.
```

Appendix B
Verbatim LLM Response

```
# Retrieval Quality in RAG Pipelines

Based on the provided excerpts, I can explain some methods
for improving retrieval quality, though the documents do
    not
provide comprehensive coverage of how quality is *measured
    *.

## Methods for Improving Retrieval Quality

**Re-ranking with Cross-Encoders:**
The excerpts indicate that retrieval precision can be
    improved
through re-ranking using cross-encoders, which score
query-chunk pairs *jointly* rather than independently. This
represents a refinement step after initial retrieval.

**Chunking Strategies:**
The documents mention that chunking strategies---including
fixed-size windows, sentence-level splits, and semantic
boundary detection---help "balance recall and context
coherence," which suggests these design choices impact
retrieval effectiveness.

**Retrieval Method Selection:**
Two complementary approaches are described:
- Dense retrieval (vector embeddings): semantic similarity
  via cosine or dot-product distance
- Sparse retrieval (BM25): a probabilistic method based on
  term frequency and inverse document frequency, noted as
  "effective for keyword-heavy queries"

## Limitations

The provided excerpts do not contain explicit information
about:
- How retrieval quality is *measured* (e.g., precision,
  recall, MRR, NDCG)
- Quantitative benchmarking approaches
- Evaluation frameworks or datasets

To fully answer your question about measurement
    methodologies,
additional source material would be needed.
```

The response is grounded strictly in the approved chunks and correctly identifies the missing measurement metrics—the chunks that would have discussed adjacent topics (hybrid fusion, hallucination) were removed by the gate, so their absence is reflected honestly rather than fabricated.

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